

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence. Where no observation exists, emem signs an absence and its reason, not a zero.

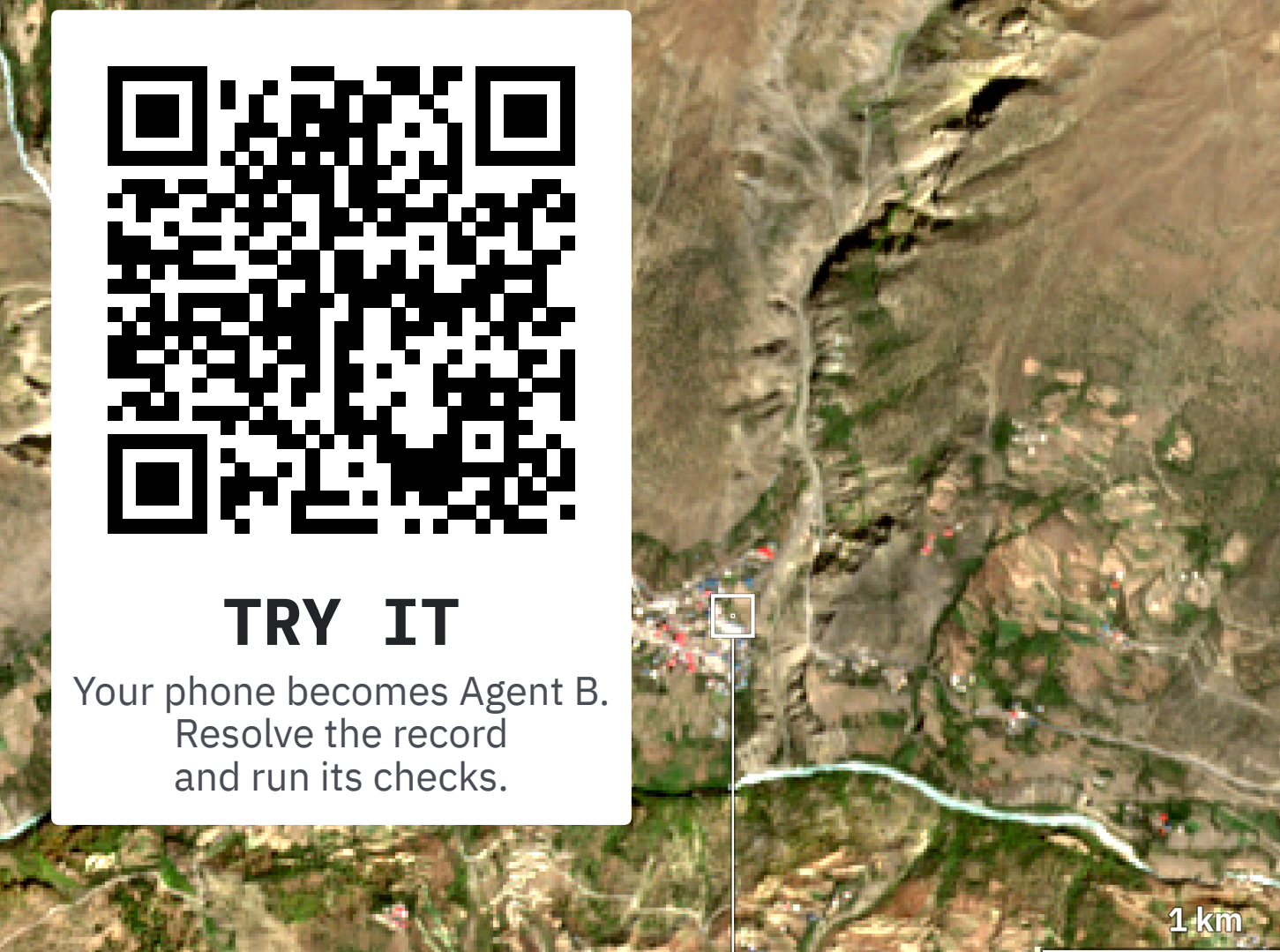
A compact citation keeps the observation traceable across agents and time. Design goal: encode in orbit, decode with AI.

Jaya Kumari · Avijeeet Singh · Vortex AI · avijeeet@vortex.ai · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026

Agent A cites a satellite observation. What can agent B check without trusting A or A's model?
emem makes satellite observations, raw or derived, addressable by place, band and time. Each is a record named by the BLAKE3 hash of its bytes and signed in a batch. The exact evidence A cites survives a change of model and a handoff; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name or signature. Trace (token) (record bytes) (cell) (pixel) (receipt) (log) (key)



TRY IT
Your phone becomes Agent B.
Resolve the record
and run its checks.



2.02 GB scene 84 characters
The reference to this pixel's NDVI record fits in one text message.
84 characters; checking it fetches the 1,115-byte record and its log entry.

From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · bind · sign · source Policy: refuse if one fails.	Reason with cited evidence Carry the reference onward.

1 Does agent B act on a corrupted handoff?

The same corruptions, five handoff conditions.

Agent A cites a satellite reading. A relay or a faulty signer changes the evidence. Agent B acts on it, declines it or acts on the genuine value.

Agent A cites

handoff

relay

Agent B

B did not act on corrupted evidence agents, pooled Claude

declined

acted on genuine value

Agent A reads NDVI 0.4709 and cites it
Sentinel-2A L2A, 25 Sep 2026
outlined: the 10 m pixel the record names
RGB from recorded grids

prose
The Sentinel-2 L2A NDVI for the 10 m cell at Keylong, Laha...
JSON
{ "about": { "field": "Keylong, Lahaul (32.57126 N, 77.03448...
retrieved text (BM25)
Field log, Keylong, Lahaul (32.57126 N, 77.03448 E), cell ...
opaque id
record ref obs-7d1c44
emem reference + verifier
emem: fact: defi. zb572. xoso. zb1ec: oj5ceccile62uvm6hedpuk6...

relay or faulty signer
one change per trial,
shared corruption families:
value · unit · cell · time ·
band · source · derivation ·
pixel · stale record ·
signature / id

Agent B
reads the text
reads the fields
retrieves a passage
fetches the sender's record
resolve re-hash place / time / band signature
log recompute source re-read

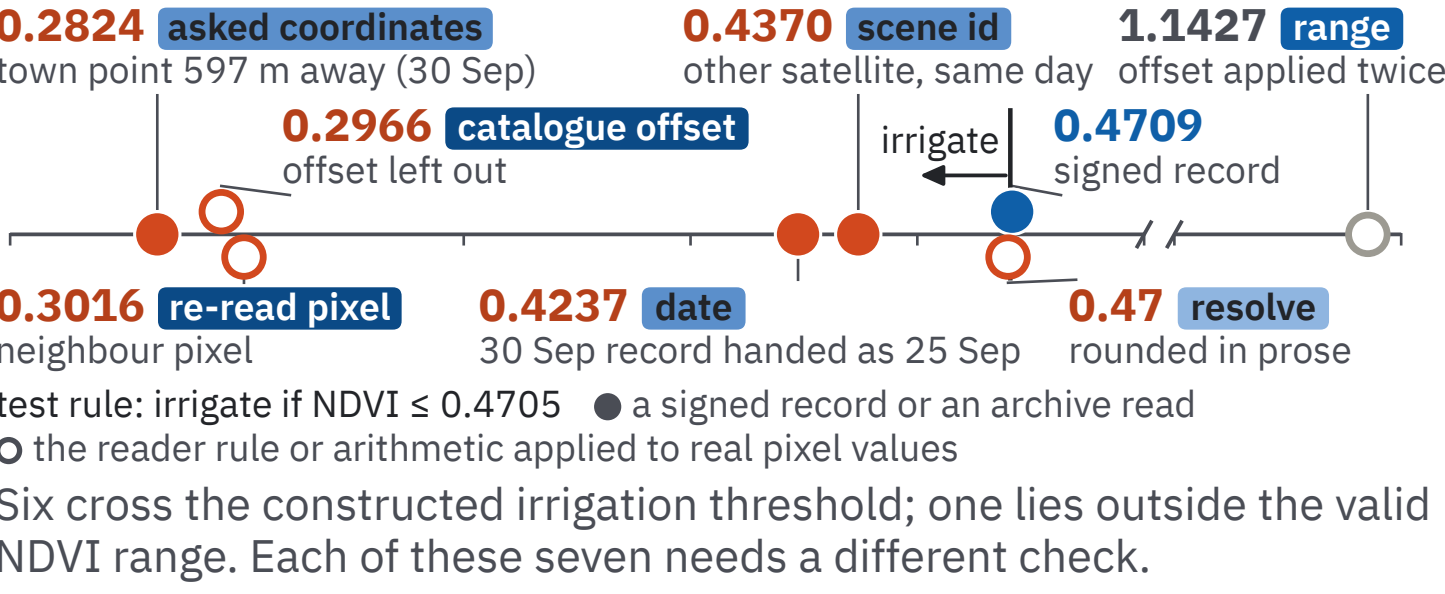
22 / 276	22	0
112 / 276	106	6
56 / 276	56	0
134 / 288	88	46
300 / 300	264	36

Matched ablation, deterministic receiver, same 23 corruptions (4 Oct), accepted: JSON, no checks 23 → emem's checks minus hash, signature and log 8 → plus four consistency checks 3 → emem's verifier 0
Across the 23 shared items, agents with an emem reference did not act on 276 of 276 corrupted handoffs (240 declined, 36 acted on the genuine value); with an opaque id, 122 of 276; in prose, 22 of 276.
By design, Earth data stays on the encoding device in orbit; its signed record and reference are downlinked, so the source is not re-read (SAT-042 harness, not flown).

2 How does a value change between agents?

One NDVI, eight values

We varied one Keylong NDVI by scene, date, place, pixel and arithmetic.



3 Why can a value change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz: change in readout. The terms name possible causes (environment, sensor, geolocation, encoder, residual); we have not measured their shares.

7 What does a checked reference establish?

The record, not physical truth

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token record source pixel

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG[row, col] = DN(b) & Ed25519(α) & leaf in STH & STHgeo.qa prefix of STH
Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.
1-3 content address 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: BLAKE3 and the verifier's code
4-5 cell binding hashed cell = token cell = the lat/long the derivation used · the cell rule
6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem's code
8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · this STAC item carries no checksum
10-11 server signature emem.dev's Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right
12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the signed tree head (STH) geo.qa co-signed · geo.qa is also run by Vortex AI
15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC
Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

EO evidence that travels between agents, checked by the receiver down to the source pixel.

Content hashes (IPFS), signed logs (SCITT) and provenance (PROV) already exist. emem adds typed EO references and a source re-read, which caught a wrong pixel that hashing, signatures, logs and recomputation had passed (panel 5).

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.

SOURCE FILES
named, not hashed
B08 B04
B04 + B08 COGS
Planetary Computer
277.9 + 281.9 MB
row 9443, col 9098
DN 1900, 3502

RECORD · 1,115 B
12 fields as decoded; no signature inside
kind cell band tslot value confidence 0.95 fixed per scene class (SCL4 here), not a measured uncertainty
sources sentinel2_2_12a ...B08_10m.tif ; ...B04_10m.tif (named): captured 2026-09-25T05:42:51Z
derivation sentinel2_12a_indices.ndvi@1 · EPSG 32643 scene S2A_MSIL2A_20260925T054251_R005_T43FS... (named): DNs 3502 / 1900 · offset -1000 · + 9 more args

ADDRESS
BLAKE3 of the record
oj5ceccile62uvm6hedpuk67cj
t2pgc7z33mts
uffgaxbxyaa
changes if one bit of the record changes
leaf of a batch root
log entries

ATTESTATION + LOG
Ed25519 over a batch root key 777er3yi...
Merkle log, RFC 6962 style, BLAKE3

emem: fact: defi. zb572. xoso. zb1ec: oj5ceccile62uvm6hedpuk67cj t2pgc7z33mts yuffgaxbxyaa

handed over: 84 characters, 46 GPT-4 tokens

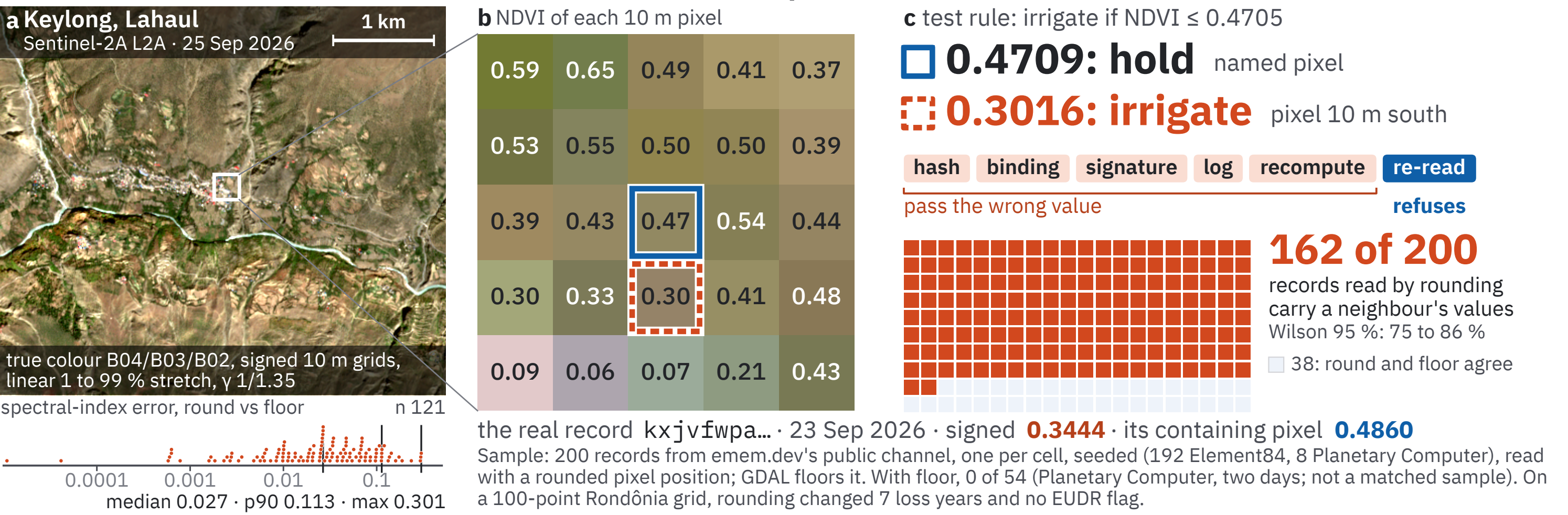
BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE RECORD, AS CODED · fact.rs, cbor.rs, geo.rs

fact f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation{fn key, args}, privacy, schema_cid, signer, signed_at)
name b = CBOR(f), declaration order, NaN as f9 7e00, -0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem: fact: <cell>: <fact_cid>
cell lat_q = round((lat+90)/180 · (2³¹-1)), lng_q = round((lng+180)/360 · (2³¹-1)); pitch 8.583e-5° = 9.52 × 8.06 m here

5 A valid record can preserve a wrong source read

NDVI = (DN₈ - DN₄) / (DN₈ + DN₄ + 2o), o = -1000 (product offset)



the real record kxjvwfpa... · 23 Sep 2026 · signed 0.3444 · its containing pixel 0.4860
Sample: 200 records from emem.dev's public channel, one per cell, seeded (192 Element84, 8 Planetary Computer), read with a rounded pixel position; GDAL floors it. With floor, 0 of 54 (Planetary Computer, two days; not a matched sample). On a 100-point Rondônia grid, rounding changed 7 loss years and no EU DR flag.

6 Which changes can the receiver detect?

Checks can continue back to the source

Resolve Re-hash Bind Signature Log Recompute Re-read

agent bars: three Claude models pooled, 1 Oct 2026

	prose	JSON	RAG (BM25)	opaque id	emem checks	first check that refuses
agent bars: three Claude models pooled, 1 Oct 2026						
seen in production						
value						
G0						
M1						
M2						
M8						
cell						
M9						
time						
M10						
band						
M6						
source						
M11						
derivation						
M12						
M14						
stale / current						
M5						
M6						
signature / id						
M7						
M13						
source pixel						
M15						
entity						
M17						

B did not act on corrupted evidence

deterministic receiver, same corruptions
hash binding signature log log recompute re-read D to I are cumulative: each adds one check.
Bars: B did not act on it (declined, or acted on the genuine value), n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5); squares: deterministic ceiling. Totals: 23 to 25 items, 1 Oct 2026. Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Each added check refuses a corruption that the checks before it pass. Agents with an emem reference decided correctly on 71 of 72 genuine controls and declined one.

Reading the fields they carry, agents did not act on 112 of 276 corrupted JSON handoffs and 134 of 288 with an opaque id. On panel 5's real record, read live, agents acted 4 of 4 times uninstructed, 2 of 4 with resolve-and-bind instructions and 0 of 4 with a source re-read.

8 Can a later agent recover the earlier state?

Revisit the earlier evidence

$$C_t = TxAsOf(O^*, a, b, t)$$

$$\text{recall} = \max_{(t, cid)} \{O \text{ in } C_t : t \leq t^*\}$$

Latest-as-of mode: take the versions signed by τ (signed_at, the signer's clock), then the latest valid time s* t*. If no candidate exists, recall returns an empty result; CID order breaks ties.

BENGALURU · SAME CELL AND BAND · 2026

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90 earlier CID	Copernicus DEM30 later CID
as of 15 Jun: 918.0 m	

The next agent can recover the earlier citation.

The later 915.07 m reflects a provider change.

9 What does the memory contain?

Observations + temporal edges

$$M = (O^*, E^*) \quad O = (a, b, t, v, u, p, s)$$

a: location cell · b: variable · t: valid time · v: value
u: uncertainty, if recorded · p: provenance, recipe, signed_at
s: attestation associated with the observation

$$E = (\text{subject, predicate, object, } t_{\text{from}}, t_{\text{to}})$$

Append observations and supersedes / disagrees_with edges; keep earlier records.

Lookup = (a, b, t); a cid names a record, not an object
cid(record) = BLAKE3(CBOR(record))

Use it in an EO workflow

Compare dates · recall, diff

Assemble evidence · merge, trace

Retain conflicts · competing, evolve

Refresh inputs · ensure, valid

Compare vegetation observations, keep conflicting estimates and pass their references on. A request reuses stored records and fetches and signs missing ones from the archive.

10 What else can an agent carry?

From a place to a device run

emem's tokens share one grammar.

data need	grammar, after emem:	what its id hashes
a place	cell: <cell164>	the quantised lat/long; no hash
one value	fact: <cell>: <cid>	BLAKE3(CBOR(fact)), 52 chars
nothing there	fact: · kind absence	same; + BLAKE3(reason)[:16]
an embedding	fact: · 128 / 1024 floats	same; not re-Runnable
a derived value	fact: · op, parents	same; pure ops re-Runnable
a raster	raster: <a>: : <t>: <d>	fact d names BLAKE3(grid)
a raster in time	cube: <a>: : <t0...t1>: <d>	BLAKE3(CBOR([member d...]))
a raster set	raster: <set>: <d>	BLAKE3(CBOR([d..., purpose]))
many facts, one id	bundle: <cid>	BLAKE3(p, cell band t cid)[:16]
a file, one chunk	tree: <file>: #row=i	BLAKE3(note)[:16]; root inside
a reasoning stage	state: <cid>	BLAKE3(CBOR(state record))
a named object	entity: <cid>	BLAKE3(anchor)[:16]: a label
a device run	trace: <cid>	BLAKE3(CBOR(signed trace))
an evidence chain	track_v1 note	link = BLAKE3(prev, ref)[:16]

14 data needs: embedding and absence share the fact grammar.

Names, sets and records hash differently. Grey: archived embeddings; device runs in a reference harness. One bundle resolves to eight facts; four reasoning-state addresses re-hash. Neither check validates an agent's reasoning.

11 Satellite execution evidence

A signed trace binds the reported run

SAT-042 scripted pass · reference harness · 30 Sep 2026.

Enrol	Write, no trace	Capture the pass	Unbound fact	Bound batch	Score vs anchor	Rewrite one log
ENROLLED	REFUSED	SIGNED	REFUSED	ADMITTED	SCORED	REFUSED
8 trace layers, each log digest chained; seq 2 rewritten after signing: broken at seq 3						
syscall	scheduler	memory	sensor bus	signal	energy	thermal
seq 0	seq 1	seq 2	seq 3	seq 4	seq 5	seq 6

Reference harness, no spacecraft enrolled. The trace shows which key signed the run and that no logged segment changed; it does not show which code ran on which inputs. The gate binds output digests, not band, cell or time.

12 Use emem. Carry the evidence forward.

ChatGPT · @emem plugin
"Signed facts about real places"

Claude · Directory plugin / MCP
"hand another agent a token it can check offline"

Diffy · Marketplace plugin
"Shared memory for AI agents. One address per fact, one signature you check."

GitHub MCP Registry · Server listing
"Satellites for AI."

ClawHub · OpenClaw plugin listing
"Shared, signed and content-addressed memory of the physical world..."

CONNECT
Setup · EO tasks
Reproduce

MORE CLIENTS
Cursor · VS Code · Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript · Docker
LangGraph memory · GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex · AutoGen
CrewAI · Mastra · Agno (see setup notes)

DISCOVERY
MCP Registry · MuleSoft Exchange · Glama · Smithery
Context7